Anamol Shrestha

Fort Worth, TX (relocating to Bay Area) • (940) 735-0096 • anamol.stha@gmail.com • linkedin.com/in/anamol-s-0b9065168

PROFESSIONAL SUMMARY

Staff-level distributed systems engineer with 8+ years building fault-tolerant, high-concurrency execution infrastructure at scale. The coordination, concurrency, and fault-tolerance challenges in LLM agent orchestration are problems I've been solving for years at a different abstraction layer - from Kafka-driven multi-step workflow execution at SEI ($2B+ daily, 100K+ events/day, exactly-once delivery) to event-driven task coordination across 50+ applications at Fidelity. Strong in Python and TypeScript; Java-primary background with deep systems-level thinking. Seeking a Staff IC role to bring production-hardened distributed systems expertise to the emerging agentic AI execution domain.

WORK EXPERIENCE

Fidelity Investments — Aug 2024 - Present

Principal Full Stack Engineer - OpsWorX Platform — Westlake, TX

- Architected the **SDK event management and task coordination framework** powering the OpsWorX interoperability platform - enabling **concurrent multi-agent workflows** across 50+ hosted applications with 40% latency reduction and 60% fewer inter-app failures.
- Led development of Java Spring Boot backend APIs serving 20+ internal and partner engineering teams with **99.9% uptime** - drove all architectural decisions and owned the full execution lifecycle from design through production.
- Drove **zero-downtime migration** of 15+ legacy WPF/Prism applications to a modern microservices-based orchestration platform, reducing deployment cycle time by 35%.
- Owned the **OpsWorX Desktop Application and TypeScript SDK** - an Electron + Angular/TypeScript platform consumed by 20+ partner teams; authored typed API contracts, versioned the SDK, and published integration documentation that cut onboarding time by **50%**.
- Built **TypeScript-based cross-application event APIs** via io.Connect - real-time data sharing, state synchronization, and global notifications with sub-50ms propagation across Fidelity's 40,000+ employee organization.
- Standardized REST API integration patterns across 5+ business units, eliminating platform fragmentation and enabling consistent TypeScript client consumption.
- Mentored 5+ engineers through design reviews and hands-on guidance; established platform-wide coding standards that accelerated feature delivery velocity by 25%.

SEI Investments — Mar 2022 - Aug 2024

Sr. Java Developer - Special Assets Platform — Remote

- Architected a fully **event-driven multi-step workflow execution system** using Kafka and JMS to orchestrate the TBA MBS trade lifecycle end-to-end - processing **100,000+ tasks daily**, replacing synchronous flows with fault-tolerant async execution and cutting settlement latency by **45%**.
- Designed **distributed task coordination** with message-key-based deduplication, transactional producers, and custom retry/recovery mechanisms - guaranteeing **exactly-once execution** at **99.95% reliability** across a multi-system pipeline.
- Developed Spring Boot REST APIs integrated with **real-time event streams**, supporting 8+ consuming services with sub-100ms response times under peak concurrent load.
- Key contributor to financial instruments platform supporting **$2B+ in daily trade volume** - owned a net-new asset class (TBA MBS) end-to-end from architecture through production, demonstrating full platform engineering ownership.

American Airlines — Aug 2018 - Mar 2022

Software Engineer - Inbound IVR Platform — Fort Worth, TX

- Maintained and modernized core Java APIs supporting **automated decision and routing workflows** handling 5M+ customer interactions monthly at 99.9% availability.
- Delivered the IVR Containment Deeplink Project - a **multi-step automated workflow** routing customers via SMS self-service, achieving 12% containment and deflecting **600,000+ calls annually** ($3M+ in operational savings).
- Enhanced the Proactive Greeting Service with **real-time personalization using concurrent event processing** - delivering 1M+ personalized responses monthly with sub-200ms latency.
- Led decomposition of Java monolith into **microservices on PCF/AWS** - improved deployment frequency by 4x and reduced production incidents by 30%.

EDUCATION

University of Cumberlands December 2022

MS, Information Technology

University of Louisiana at Monroe April 2018

BS, Computer Science | BS, Mathematics

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C#, SQL, Shell Script, Groovy

Distributed Systems & Orchestration: Microservices, Event-Driven Architecture, Multi-Step Workflow Execution, Concurrent Task Coordination, Fault-Tolerant Pipelines, DDD, CQRS, Saga Pattern, High-Concurrency Low-Latency Systems

Messaging & Streaming: Kafka, Kafka Streams, JMS, RabbitMQ, ActiveMQ - exactly-once delivery, transactional producers, distributed deduplication

Backend & Frameworks: Spring Boot, Spring Cloud, Node.js, Express.js, gRPC, REST, GraphQL

Cloud & Infrastructure: AWS (EC2, S3, Lambda, SQS, SNS, RDS), Azure, GCP, Docker, Kubernetes, Helm, Terraform, Istio, PCF

Databases: PostgreSQL, Oracle, MySQL, MongoDB, Redis, Elasticsearch, DynamoDB

Observability & CI/CD: Datadog, Dynatrace, Prometheus, Grafana, ELK Stack, Splunk, Jenkins, GitHub Actions, ArgoCD

AI & Tooling: Claude Code, GitHub Copilot, Cursor, Prompt Engineering, LLM API integration